# Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

### Division B | Solar System | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

#### 1. (2 points) Which three broad needs define habitability for life as we know it?

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

A. Oxygen, sunlight, and soil

B. A magnetic field, rings, and moons

C. Ice, methane, and nitrogen only

D. Liquid water, energy, and key chemical elements

**Correct: D**

#### 2. (2 points) A planet moves from 1 AU to 2 AU. By what factor does the starlight flux change?

Unit: Light, radiation, and temperature.
Worked-case data: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth's 255 K.
Separate investigation observations:
Distance: A planet orbits a Sun-like star at 0.72 AU.
Albedo: Very high: about 0.75, from bright clouds.
Predicted Teq: About 230 K, below freezing.
Measured surface: About 735 K.

A. It becomes 1/8 as large

B. It becomes 1/4 as large

C. It becomes 1/2 as large

D. It doubles

**Correct: B**

### 3. (2 points) A body orbits the Sun at 9 AU. What is its period?

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

A. 9 years

B. 81 years

C. 3 years

D. 27 years

**Correct: D**

### 4. (2 points) Why is Venus's surface hotter than Mercury's?

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

A. Venus has active tidal heating

B. A thick $CO_2$ atmosphere creates a runaway greenhouse

C. Venus is closer to the Sun

D. Venus has a strong magnetic field

**Correct: B**

**5. (2 points) Which moon has its own internally generated magnetic field?**

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

A. Europa

B. Titan

C. Enceladus

D. Ganymede

**Correct: D**

**6. (2 points) Which mission returned a sample of asteroid Bennu?**

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

A. Kepler

B. OSIRIS-REx

C. Rosetta

D. Dragonfly

**Correct: B**

**7. (2 points) Which method directly gives a planet's radius?**

Unit: Exoplanets: detection and the named systems.
Worked-case data: An Earth-sized planet transits two stars: a Sun-like star and a red dwarf with one-tenth the Sun's radius. Compare the transit depths.
Separate investigation observations:
Light curve: Seven different periodic dips in brightness.
Star: An ultracool red dwarf, about 40 light-years away.
Sizes: All planets are roughly Earth-sized.
JWST: The two innermost planets show no thick atmosphere.

A. Radial velocity

B. Microlensing alone

C. Astrometry

D. Transit

**Correct: D**

**8. (2 points) Why are small, hot worlds poor at keeping light gases?**

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

A. Gravity is too strong

B. Light gases freeze out

C. The Sun adds gas

D. Gas molecules move fast compared with a low escape velocity

**Correct: D**

**9. (2 points) If a star’s surface temperature doubles, its emitted power per area changes by:**

Unit: Light, radiation, and temperature.
Worked-case data: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth’s 255 K.
Separate investigation observations:
Distance: A planet orbits a Sun-like star at 0.72 AU.
Albedo: Very high: about 0.75, from bright clouds.
Predicted Teq: About 230 K, below freezing.
Measured surface: About 735 K.

A. 8 times

B. 16 times

C. 2 times

D. 4 times

**Correct: B**

**10. (2 points) If the distance between two masses triples, the gravitational force becomes:**

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth’s mass and 2 times Earth’s radius. Compare its surface gravity and escape velocity with Earth’s.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

A. 1/3 as large

B. 3 times as large

C. 1/27 as large

D. 1/9 as large

**Correct: D**

**11. (2 points) In a simplified habitable-zone model, stellar luminosity increases by a factor of four. By what factor must distance increase to preserve incident flux?**

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , .

**12. (2 points) A synthetic planet and star have radius ratio 1/100. For a uniform stellar disk and central transit, what fraction of light is blocked?**

answer: 0.0001 (2 points).

Numerical target 0.0001; absolute tolerance 2e-07. Accepted units: , .

**13. (2 points) Compare surface-water and subsurface-water habitability arguments using Europa or Enceladus as listed examples.**

answer: Surface conditions may be hostile while subsurface liquid can be maintained by internal/tidal heating. Liquid, chemistry and usable energy must all be considered; water alone does not prove life. (2 points).

**14. (2 points) Titan has surface liquid hydrocarbons and water ice. Explain why habitability questions should specify the solvent and biological assumptions.**

answer: Habitability depends on the proposed chemistry and solvent; the presence of one kind of liquid does not automatically imply conditions suitable for familiar water-based life. (2 points).

**15. (2 points) A spectrum from a Mars teaching dataset shows a mineral associated with past water alteration. What can and cannot be concluded?**

answer: It can support a history of water-related processes under suitable geological context, but does not establish that life existed or that stable liquid water is present now. (2 points).

**16. (2 points) Explain the difference between evidence for organic molecules on Bennu or comet 67P and proof of biology.**

answer: Organic molecules can form through non-biological processes; their presence constrains chemistry and delivery pathways but is not by itself evidence of life. (2 points).

**17. (2 points) Why can a planet in the TRAPPIST-1 system require atmospheric-retention evidence in addition to an estimated equilibrium temperature?**

answer: An equilibrium temperature does not specify an atmosphere, greenhouse effects, stellar irradiation history or surface pressure; these affect stable liquids and habitability. (2 points).

**18. (2 points) A planet is detected through periodic stellar radial-velocity shifts. What physical interaction produces the signal?**

answer: The planet and star orbit their shared center of mass, creating a line-of-sight stellar velocity variation; other stellar variability must be considered. (2 points).

**19. (2 points) Explain one engineering tradeoff between high spatial resolution and broad repeated mapping for an allowed mission.**

answer: Finer sampling can require narrower swaths, greater data volume or observing time; a mission must balance resolution, coverage and revisit against its science goals. (2 points).

**20. (2 points) Why is chemical disequilibrium potentially relevant to habitability but not a unique biosignature?**

answer: A disequilibrium can provide usable free energy or motivate biological hypotheses, but geological, photochemical and transport processes can also maintain it. (2 points).